

Roshan Raj

Curriculum Vitae

kaalkrit.github.io | roshan.raj@studium.uni-hamburg.de

Research Interests

Interested in exploring the conceptual foundations of quantum information science, particularly in quantum algorithms, quantum error correction, quantum communication, and quantum metrology.

Education

M.Sc. Physics Oct 2025 - Present

Universität Hamburg (UHH), Hamburg, Germany

Integrated M.Sc. Physics (5-year) 2019 - 2024

National Institute of Technology (NIT), Surat, India CGPA: 9.37 / 10.0

Thesis: *Quantum and Classical Applications of the Relativistic Relationship of Kinetic Energy and Momentum*. Guide: Dr. Vikash K. Ojha.

Briefly enrolled: PhD Physics, IISER Mohali (2025); B.Sc. Physics, Patliputra University, Patna (2018).

All India Higher Secondary (CBSE) Loyola High School, Patna 2018 | 77.2%

All India Secondary (ICSE) St. Paul's High School, Patna 2016 | 91.6%

Research Experience

Lie-Theoretical studies of nonlinear Interferometry 2026 - Present

Institute for Quantum Physics (IQP), UHH & Supervisor: Dr. Schlawin

- A student project on $\mathfrak{su}(1,1)$ interferometry and the possibility to generalise to $\mathfrak{su}(p,q)$ to investigate quantum sensing limits.

Integrated M.Sc. Thesis 2023 - 2024

NIT Surat & Supervisor: Dr. Ojha

- Tackled the relativistic anharmonic quantum harmonic oscillator and Maxwell-Jüttner distribution under a novel relativistic formalism.

Selected coursework

At UHH: Quantum Computing and Information I & II (Prof. Dieter Jaksch) · Light fields & Quantum Correlations (Prof. Roman Schnabel) · Quantum Technology I & II (Prof. Ludwig Mathey) · Quantum Metrology and Sensing (Dr. Frank Schlawin) · Quantum Statistics with path integral (Prof. Michael Thorwart)

At NIT: Computational Methods · Experimental Techniques · Mathematical Physics · Quantum Field Theory

Earlier Projects

- Reviewing observables in classical and quantum mechanics. osf.io/p2ufx (2023).
- An open-ended story on quantization. osf.io/92bvp (2023).
- On the investigation of two non-neutral static bodies. [arXiv.2209.10641](https://arxiv.org/abs/2209.10641) (2022).
- The pictures of quantum dynamics. Unpublished Book chapter (Hindi) (2022).
- Dynamics of two objects considering the minimum total potential energy principle. osf.io/ukx9t (2021).

Schools, Seminars & Workshops

- European Quantum Technology Summer School, AUAS Amsterdam (2026)
- ICTP PWF Bangladesh: School on Quantum Field Theory, ICTP Italy (Online, 2024).
- Short Course in Math History - Prof. A. Viramani, IIT Gandhinagar (2021).
- National Workshops on MATLAB and Data Analysis, BBD NITM Lucknow (2020).

Achievements & Qualifications

- Joint CSIR-UGC National Eligibility Test (NET), Physical Sciences - Qualified (2025)
- National Talent Search Examination (NTSE) - National Merit Scholarship, New Delhi (2016-24).

Responsibilities & Positions

- Appointed Joint Academic Affairs Secretary, Academic Affairs Council, NIT Surat (2022).
- Co-Convenor, Society for Cultivation of Science and Humanities (SCOSH), NIT Surat (2021).
- Junior Graphic Designer, SCOSH Student Chapter, NIT Surat (2020).

Skills

Technical: Python (intermediate) · MATLAB · $\text{\LaTeX}$

Languages: Hindi (native) · English (proficient)

Other

- Self-published poetry collections: *KaalSangharsh* (2021), *Mano Vighn* (2023).
- Notes: ‘A Note on Consciousness’ (2026) · ‘On Einstein’s Theory of Gravitation’ (2025) · ‘On One of the Applications of Electrodynamics and Crystals’ (2025).

Declaration: *I hereby declare that the information given above is true to the best of my knowledge.*

Roshan Raj

(Roshan Raj)

Date: 05/06/2026.